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CS 360 Principles of Database Systems

Homework 2: Relational Algebra

Name:

Instructions: Print this assignment using single-side pages. Fill in your name above, and write in the solutions in the space provided below each question. You are allowed to use the back of each page. If you used any scratch paper to show your work, append those to the end. **Note:** It is important you use this format for gradescope.

Submission: After you've filled in the answers, scan all pages into a PDF, and submit to canvas.

Problems

1. Design a *schema diagram* for a company like Netflix that minimizes redundancy and NULL values. You should recall that an attribute cannot hold a list of values. Indicate all primary keys (underlined) and foreign key constraints (directed edges). You need to store data on the following:

   - A Movie has a title, description, country of origin, and a release-year. Assume that there may be two movies with the same name (e.g., Godzilla), but they are never released in the same year. You may assign each movie a unique identifier, but you shouldn't need to.
   - An Actor has a name and a biography. Assume that there may be two actors with the same name. You may assign each actor a unique identifier.
   - A User has a userid and password. The company does not allow two different users to share the same userid.
   - You need to keep track of what movies the users have seen.
   - Users can rate movies from 1 to 5 stars. Note that they need not have seen the movie to rate it! A user can only rate a movie once (though that rating can change later).
   - Users can also comment on each actor and/or movie. A comment has a date, time, and the comment itself. Unlike ratings, users can leave a multiple comments on the same movie or actor.
   - Actors play certain roles in certain movies. Actors may also play multiple roles in the same movie (see: Eddie Murphy and Mike Myers).

Space for your schema diagram is given on the next page.

2. Let's define some new relational operations using a composition of various *primitive* relational and set operators! Because we already defined $\bowtie$ operation using only primitive operators in class, you may safely use $\bowtie$ in helping define the two operators below. For the problems below, assume we have relations $R(A_1, ..., A_n)$ and $S(B_1, ..., B_m)$.

   (a) The Left Semi-Join $R \ltimes S$ retains all tuples in R that participate in the natural join between R and S. For instance, the Left semi-join of R and S as defined in the previous problem produces $\{(1, 2)\}$.

   I'll do this one: $R \ltimes S = \Pi_{A_1,...,A_n}(R \bowtie S)$. We start by producing the natural join, which by definition would return a table containing attributes $\{A_1, ..., A_n\} \cup \{B_1, ..., B_m\}$ and the concatenation of tuples in R and S that agree on their common attributes. Next, we project only the attributes $A_1, ..., A_n$ to retain the tuples in R that participated in the join.

   (b) The Symmetric Difference $R \triangle S$ returns the set of tuples found in either R or S but not both. For instance, the symmetric difference between R and S as defined in the previous problem gives us $\{(1, 2), (3, 4), (2, 2), (5, 1)\}$.

   $R \triangle S = ?$

   (c) The Right Anti-Join $R \lhd S$ retains all tuples in S that do not participate in the natural join between R and S. For instance, the right-anti join between R and S as defined in the previous problem produces $\{(5, 1)\}$.

   $R \lhd S = ?$

For the remaining problems, consider the relation instances below that model competitions in the world of college sports. Assume that each city may have multiple schools (*e.g.*, both Columbia and NYU are in New York). Schools are split into two conferences: A and B. Within each conference, there are four divisions: North, South, East, and West. Games are played by "home" and "away" (*i.e.*, the traveling) schools, identified by the schoolIDs. For instance, game 7 was between GaTech (24 points) and ASU (6 points) on 9/8/1980.

School

schoolID	city	school	conference	division	spent
0	Seattle	Washington	B	West	500M
1	Cleveland	Case	A	North	100M
2	Pittsburgh	CMU	A	North	-50M
3	San Francisco	Berkeley	B	West	800M
4	Oakland	Mills	A	West	350M
5	Tempe	ASU	A	West	400M
6	Miami	Miami	A	East	200M
7	Houston	Rice	B	South	250M
8	New York	Columbia	A	East	800M
9	New York	NYU	A	East	400M
10	Buffalo	SUNY	B	East	100M
11	Atlanta	GaTech	A	South	200M
12	Baltimore	JHU	A	North	0M
13	Lafayette	Purdue	A	South	90M

Game

gameID	Away	Home	date	year	awayScore	homeScore
0	0	11	9/2	2012	0	3
1	3	13	10/17	2009	23	0
2	5	10	10/10	2012	10	10
3	4	9	11/20	2015	17	7
4	2	7	9/27	2014	7	14
5	9	8	10/30	1990	14	15
6	8	3	8/30	1980	21	9
7	11	5	9/8	1980	24	6
8	4	2	10/28	1981	35	15
9	12	10	11/27	2012	3	40
...	...	...	...	...	...	...

Coach

name	schoolID	title
Carroll	0	Head Coach
Jackson	1	Head Coach
Tomlin	2	Head Coach
Kelly	3	Assistant Coach
Brown	1	Head Coach
Day	0	Assistant Coach
…	...	...

3. Write a relational algebra expression that lists all schools (city and school name) in the Eastern and Northern divisions of Conference "A."

4. List all 2015 games (gameID, awayScore, homeScore) in which the home team lost.

5. List all the schools (city, school) for which 'Kelly' has worked as an assistant coach. You do not know the rest of the data, so you can't simply assume that it's schoolID 3. You need to join **Coach** and **School**.

6. List all games (gameID, date, year) in which 'Columbia' participated. Assume you do not know Columbia's schoolID off-hand, so you must first find it in the School table.

7. How many of each of kind of coach does each school employ? For example, in the data that is available, Washington has 1 head coach (Caroll) and 1 assistant coach (Day). Your result should contain three columns: the school, the coach title, and the number of coaches having that title.

8. The league commissioner suspects that conference B may be accepting improper funds. Return a list of schools (city, school name) outside of conference B that spent less than than the average spent by all schools in conference B. (**Stop and think:** How can you use ‘$<$’ to compare with the average you extracted when ‘$<$’ is not a set/relational operator? When have you seen ‘$<$’ being used properly in our examples?)

9. Retrieve all games (all columns from the Game table) that occurred between JHU and Purdue from 2000 to 2010. Do not assume that you know what either school's schoolIDs are. 🌶 🌶

10. What's the number of 'Assistant Coaches' for each school spending more than $100M? Your result should contain two columns: the school's name and a corresponding count. 🌶 🌶

11. How many total points (away and home combined) were scored in the 2023 season? You are reminded that duplicate tuples are removed, which can screw up your sum! If you are applying a set operator (e.g., a union) between two columns with different names, assume that the result column is unnamed.

12. Which year(s) did the school *'Case'* record its fewest wins? Note that there may be multiple years during which *'Case'* recorded their lowest win total. Do not assume you know what Case's schoolID is. (This one may be a bit long – I am so sorry lol)